

Homework 2 – VAR with Economic Policy Uncertainty

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Part I

1. Dataset Construction

We use four monthly U.S. series from 1985M1 to 2023M3:

- **EPU**: News-Based Policy Uncertainty Index in levels, from policyuncertainty.com
- **PAYEMS**: Nonfarm payroll employment, seasonally adjusted, converted to month-on-month growth rates: $100 \times (X_t/X_{t-1} - 1)$
- **INDPRO**: Industrial production index, seasonally adjusted, same transformation
- **PCEPI**: PCE price index, seasonally adjusted, same transformation

After taking growth rates, the usable sample starts at 1985M2 (458 observations).

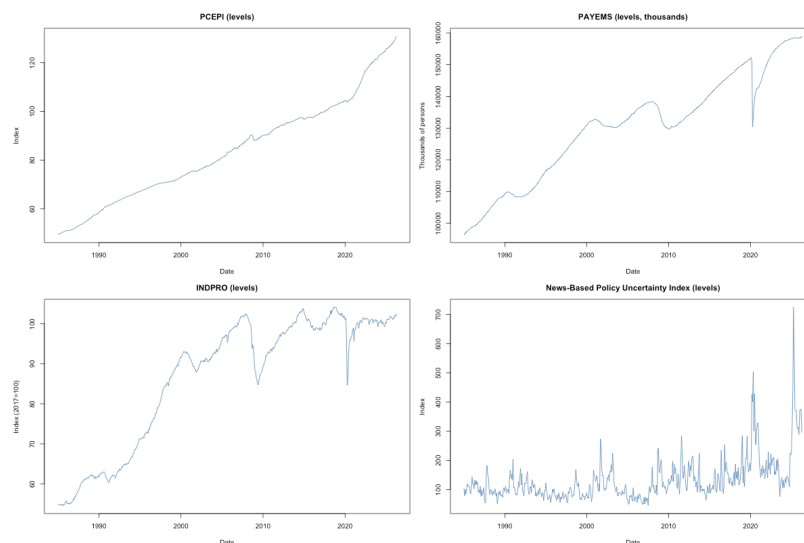


Figure 1: The four panels show the raw level series from 1985M1 to 2023M3. Top left: PCEPI, with a steady upward trend and some acceleration after 2020. Top right: PAYEMS (thousands of persons), with long-run employment growth and clear drops during the 2008–2009 recession and the COVID-19 shock in 2020. Bottom left: INDPRO, with similar cyclical patterns. Bottom right: the EPU index in levels, with high volatility and a large spike around the COVID-19 pandemic. PAYEMS, INDPRO, and PCEPI from FRED; EPU from policyuncertainty.com.

2. Reduced-Form VAR(1)

We estimate a VAR(1) on 1985M2–2019M12 (418 observations), stopping before 2020 to avoid having the COVID-19 period distort the estimated dynamics. The variable ordering is: EPU, PAYEMS growth, INDPRO growth, PCEPI growth.

2d. Estimated equations

$$\begin{aligned} \text{epu}_t = & 46.067 + 0.641 \text{epu}_{t-1} - 19.313 \text{payems_g}_{t-1} \\ & - 3.540 \text{indpro_g}_{t-1} - 13.225 \text{pcepi_g}_{t-1} + u_{\text{epu},t} \end{aligned}$$

$$\begin{aligned} \text{payems_g}_t = & 0.058 - 0.000278 \text{epu}_{t-1} + 0.679 \text{payems_g}_{t-1} \\ & + 0.017 \text{indpro_g}_{t-1} + 0.033 \text{pcepi_g}_{t-1} + u_{\text{payems},t} \end{aligned}$$

$$\begin{aligned} \text{indpro_g}_t = & 0.111 - 0.001 \text{epu}_{t-1} + 0.925 \text{payems_g}_{t-1} \\ & + 0.073 \text{indpro_g}_{t-1} + 0.458 \text{pcepi_g}_{t-1} + u_{\text{indpro},t} \end{aligned}$$

$$\begin{aligned} \text{pcepi_g}_t = & 0.168 - 0.001 \text{epu}_{t-1} + 0.021 \text{payems_g}_{t-1} \\ & + 0.009 \text{indpro_g}_{t-1} + 0.404 \text{pcepi_g}_{t-1} + u_{\text{pcepi},t} \end{aligned}$$

2e. Covariance matrix of residuals

	epu	payems_g	indpro_g	pcepi_g
epu	964.407	-0.182	-0.946	-0.098
payems_g	-0.182	0.011	0.019	0.001
indpro_g	-0.946	0.019	0.317	-0.002
pcepi_g	-0.098	0.001	-0.002	0.028

3. Impulse Response Functions

We use recursive (Cholesky) identification with EPU ordered first, meaning EPU shocks can affect all variables on impact, but no other shock moves EPU contemporaneously.

PAYEMS: Employment falls in response to an EPU shock, but relatively slowly. The trough is around month 4–5 at roughly -0.013 percentage points, and the effect is significant from months 2 to 12. The slow adjustment reflects the costs of changing workforce size – firms are reluctant to cut jobs quickly even when uncertainty rises.

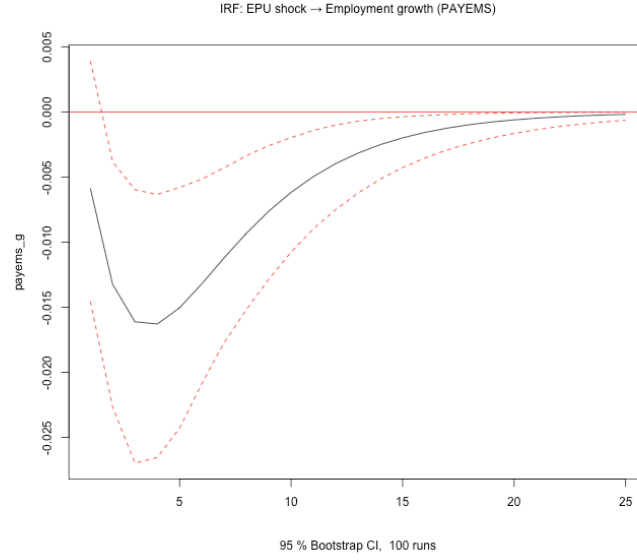


Figure 2: Impulse response of month-on-month employment growth (PAYEMS) to a one-unit structural EPU shock, from the recursive VAR(1) with EPU ordered first. Solid line: point estimate. Dashed lines: 95% bootstrap confidence bands (100 replications). Horizontal axis: months after the shock (0–24). Vertical axis: response in percentage points. Estimated on 1985M2–2019M12.

INDPRO: Industrial production reacts faster and more sharply, dropping immediately and bottoming out around month 3–4 at roughly -0.05 percentage points. The effect is significant from months 2 to 10 and fades by around month 20. Firms can cut output more easily than they can let workers go.

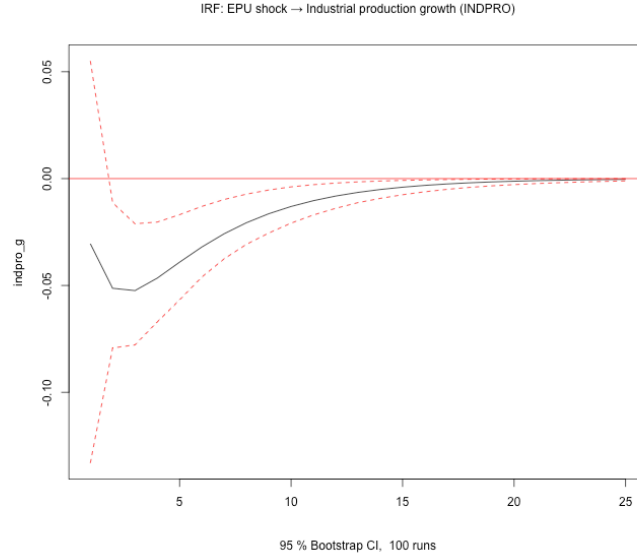


Figure 3: Impulse response of month-on-month industrial production growth (INDPRO) to a one-unit structural EPU shock, from the recursive VAR(1) with EPU ordered first. Solid line: point estimate. Dashed lines: 95% bootstrap confidence bands (100 replications). Horizontal axis: months after the shock (0–24). Vertical axis: response in percentage points. Estimated on 1985M2–2019M12.

Both responses are consistent with the standard uncertainty channel: higher policy uncertainty leads firms to cut production and delay hiring.

Part II

1. Unconditional Forecasts

Using the same VAR estimated on 1985M2–2019M12, we produce unconditional 12-month-ahead forecasts with March 2023 as the forecast origin. Expected future residuals are set to zero, so the forecast traces out the model’s implied dynamics from the current state of the system.

PAYEMS: Employment growth stays positive over the whole horizon. It starts at +0.043% in April 2023, is roughly flat in May, then rises gradually to +0.097% by March 2024. The model pulls growth back toward its long-run average throughout.

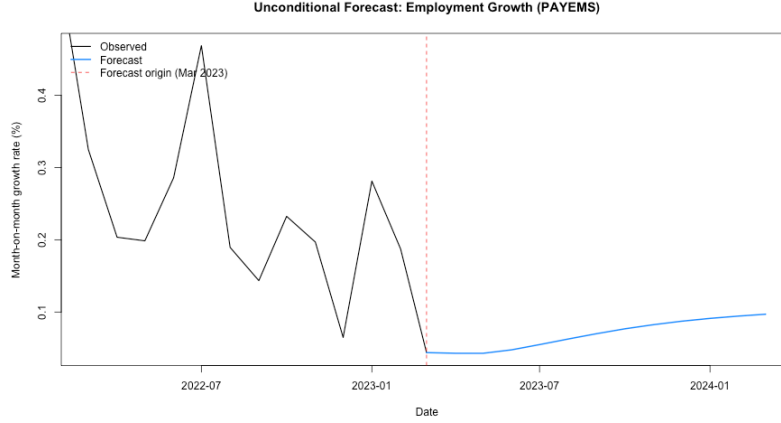


Figure 4: Unconditional 12-month-ahead forecast for month-on-month employment growth (PAYEMS), from the VAR(1) estimated on 1985M2–2019M12 with March 2023 as the forecast origin. Black line: observed values in the 12 months prior to the forecast origin. Blue line: unconditional VAR forecast. Red dashed vertical line: forecast origin (March 2023). Future residuals set to zero. Grey dotted horizontal line marks zero growth.

INDPRO: Industrial production growth starts slightly negative (-0.029% in April, -0.031% in May), turns positive in July ($+0.020\%$), and reaches $+0.126\%$ by March 2024. The brief initial dip reflects the dynamics at the forecast origin before mean reversion takes over.

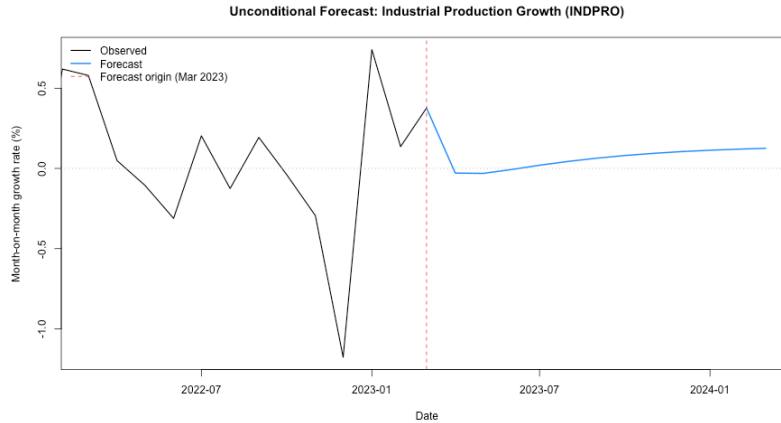


Figure 5: Unconditional 12-month-ahead forecast for month-on-month industrial production growth (INDPRO), from the VAR(1) estimated on 1985M2–2019M12 with March 2023 as the forecast origin. Black line: observed values in the 12 months prior to the forecast origin. Blue line: unconditional VAR forecast. Red dashed vertical line: forecast origin (March 2023). Future residuals set to zero. Grey dotted horizontal line marks zero growth.

Both series rise gradually over the forecast horizon, consistent with the VAR's mean-reversion dynamics toward positive long-run growth rates.

2. Structural Forecast (Election EPU Scenario)

We impose a scenario where EPU in April–August 2023 follows the same path as in October 2020–February 2021, the period around the last U.S. national elections. The imposed EPU values are:

Month (2023)	Source period	EPU value
April	Oct 2020	305.8
May	Nov 2020	327.6
June	Dec 2020	326.6
July	Jan 2021	261.3
August	Feb 2021	187.5

For each of these five months, we compute the gap between the imposed EPU value and the unconditional VAR forecast for EPU, and attribute this entire gap to an exogenous EPU shock. The implied effect on the other variables is then recovered using the first column of the normalized Cholesky impact matrix, which gives the contemporaneous response of each variable to a unit EPU shock. From September 2023 onward, EPU evolves endogenously under the VAR dynamics.

PAYEMS: Employment growth turns negative from May through November 2023, with a trough of -0.091% in July. It recovers to positive only in December 2023 and reaches +0.060% by March 2024, still well below the unconditional forecast.

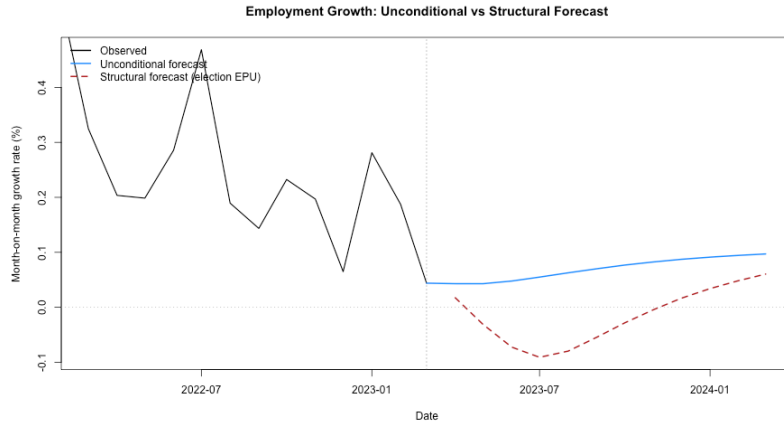


Figure 6: Month-on-month employment growth (PAYEMS) forecasts: unconditional (blue solid) vs. structural election scenario (red dashed), from the VAR(1) estimated on 1985M2–2019M12 with March 2023 as the forecast origin. Black line: observed values in the 12 months prior to the forecast origin. Grey dotted vertical line: forecast origin (March 2023). The structural forecast imposes the EPU path from October 2020–February 2021 for April–August 2023, attributing EPU deviations from the baseline to identified EPU shocks via the Cholesky impact matrix. From September 2023, EPU is endogenous.

INDPRO: The drop is much larger. Industrial production falls to around -0.446% in June–July 2023, stays deeply negative through November, and barely turns positive (+0.051%) by March

2024.

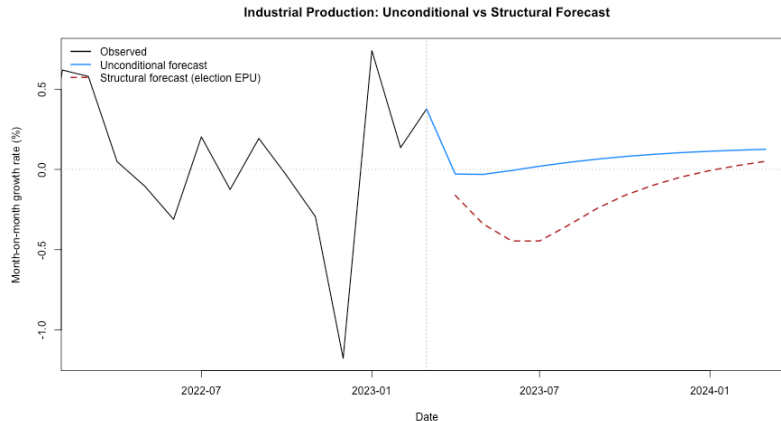


Figure 7: Month-on-month industrial production growth (INDPRO) forecasts: unconditional (blue solid) vs. structural election scenario (red dashed), from the VAR(1) estimated on 1985M2–2019M12 with March 2023 as the forecast origin. Black line: observed values in the 12 months prior to the forecast origin. Grey dotted vertical line: forecast origin (March 2023). The structural forecast imposes the EPU path from October 2020–February 2021 for April–August 2023, attributing EPU deviations from the baseline to identified EPU shocks via the Cholesky impact matrix. From September 2023, EPU is endogenous.

3. Comparison of Forecasts

The two forecasts differ considerably. The unconditional forecast shows mild positive growth for both series over the whole horizon. The structural forecast shows a sharp contraction in mid-2023, with employment and industrial production both falling well below zero before recovering slowly.

The key difference is how EPU enters the forecast. In the unconditional case, EPU evolves on its own based on past dynamics, producing a modest path. In the structural case, we force EPU onto the election scenario and treat the entire gap relative to the baseline as exogenous EPU shocks. These shocks feed through the Cholesky impact matrix and hit all variables contemporaneously, producing a much larger and more persistent contraction.

The structural forecast answers a specific question: if the spike in EPU were entirely driven by an exogenous structural shock of the kind the model identifies, how much would output and employment fall? Rather than just conditioning on EPU being high, it attributes that high EPU to a specific structural disturbance and traces its propagation through the economy. The answer is a significant and prolonged contraction, especially in industrial production.

4. Three Shortcomings

- **The VAR is small.** With only four variables, the model leaves out things like interest rates, credit conditions, or exchange rates that could matter for how uncertainty shocks transmit. This risks omitted variable bias in both the IRFs and the forecasts.
- **Recursive identification may not hold.** The Cholesky ordering assumes EPU does not react to employment, output, or inflation surprises within the same month. In practice, policy uncertainty can move quickly in response to macro news, which would make the identifying assumption invalid and distort the estimated impulse responses.
- **The EPU scenario is not a clean election signal.** The October 2020–February 2021 period was also heavily shaped by the COVID-19 pandemic. Using EPU values from that period to simulate a hypothetical 2023 election mixes in pandemic-related uncertainty, not just electoral uncertainty, which limits what we can say about the effect of elections specifically.